IS507: Data, Statistical Models, & Information
Project Report
Credit Risk Analysis: Assessing ML Models & Identifying Risks

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Abstract

This research investigates the application of diverse machine learning (ML) algorithms to predict individual credit risk and analyze the factors influencing credit behavior. We employed advanced ML techniques, including Gradient Boosting, Neural Networks, and Random Forests, to model and predict credit risk, while performing a comparative evaluation of these models based on accuracy and interpretability.

The study utilizes an anonymized dataset comprising demographic details, credit behavior, and financial background of customers. Following extensive data preprocessing, each model was trained and tested to identify the best-performing algorithm. Among the models evaluated, Neural Networks achieved the highest predictive accuracy, demonstrating their superior capability for this task. However, Gradient Boosting and Random Forests also exhibited significant potential, particularly in terms of interpretability and robustness.

The findings of this study contribute to the ongoing development of ML-driven solutions in the credit risk domain. Future research will focus on enhancing model complexity, incorporating a broader range of algorithms, and integrating non-financial datasets to address more intricate challenges in financial risk management. This work underscores the transformative potential of machine learning in improving decision-making in credit risk assessment.

1. Introduction

In modern economies, credit is essential for financing both personal and professional endeavors, enabling individuals and businesses to invest in opportunities that would otherwise be unattainable. However, as credit usage continues to rise, so does the risk of defaults, making effective credit risk assessment a crucial aspect of financial decision-making. Traditional methods of assessing creditworthiness often rely on simplified models that fail to account for the complexity of borrower behaviors and financial profiles. In light of this, contemporary machine learning (ML) techniques have gained prominence due to their ability to analyze large and intricate datasets, detect non-linear patterns, and provide more accurate predictions of credit risk.

This project aims to explore the potential of advanced ML models in predicting credit defaults and assessing customer creditworthiness. Specifically, we conduct a comparative analysis of traditional credit scoring models and modern machine learning techniques, to identify which models offer the best predictive accuracy and interpretability. Additionally, this study will delve into the various factors that contribute to credit risk, such as demographic information, financial behavior, and repayment history, to provide a more nuanced understanding of what influences a borrower's likelihood of default.

The primary objectives of this project are as follows:

- To develop a robust machine learning framework for predicting credit defaults, leveraging contemporary algorithms such as gradient boosting, neural networks, and random forests.
- To compare the performance of traditional credit risk models with newer machine learning techniques, evaluating them based on accuracy, interpretability, and scalability.
- To identify and analyze the key contributing factors influencing credit risk, which can be used to refine predictive models and inform credit assessment strategies.

By conducting this research, we aim to enhance the understanding of credit risk modeling, potentially leading to more reliable and inclusive credit scoring systems that can better predict defaults and help financial institutions make more informed lending decisions.

2. Related Work:

To understand the evolution of credit risk assessment, we first examine foundational works in the field. Galindo and Tamayo (2000), in *"Credit Risk Assessment Using Statistical and Machine Learning: Basic Methodology and Risk Modeling Applications,"* laid the groundwork by using traditional statistical models like logistic regression, which proved effective in categorizing creditworthiness. However, they acknowledged the limitations of these methods in capturing the complex, non-linear relationships inherent in borrower data, prompting a shift toward more advanced machine learning (ML) approaches.

Building on this, Addo, Guegan, and Hassani (2018) in *"Credit Risk Analysis Using Machine and Deep Learning Models"* introduced more advanced ML techniques, such as gradient boosting and neural networks, which outperform traditional methods by handling large and complex datasets. These models demonstrated enhanced accuracy in recognizing intricate patterns but also raised concerns regarding interpretability and transparency, particularly in real-time credit risk assessments. Mhlanga (2021), in *"Financial Inclusion in Emerging Economies: The Application of Machine Learning and Artificial Intelligence in Credit Risk Assessment,"* further expanded the scope by exploring the use of AI and ML to improve financial inclusion, especially in underserved populations. By leveraging alternative data sources like mobile transactions and social media activity, AI-driven credit scoring models provide a broader view of applicants' risk profiles, addressing gaps in traditional credit assessment models.

Recent works have continued to push the boundaries of credit risk prediction. A systematic review by Noriega et al. (2024) in *"Machine Learning for Credit Risk Prediction: A Systematic Literature Review"* evaluates the effectiveness of various ML models, with a focus on boosted models. They highlight both the strengths of these techniques in predictive accuracy and the challenges they face, such as data imbalance and the need for enhanced explainability. Additionally, Emmanuel et al. (2024), in *"A Machine Learning-Based Credit Risk Prediction Engine System Using a Stacked Classifier and a Filter-Based Feature Selection Method,"* proposed a stacked classifier approach, integrating multiple models to improve prediction accuracy. Their results demonstrated the potential of feature selection techniques to optimize model performance, with strong results across diverse datasets, further illustrating the growing capability of ML models in handling complex credit risk scenarios. These studies collectively emphasize the transition from traditional, rule-based scoring to more sophisticated AI-powered models that offer greater accuracy, inclusivity, and adaptability in assessing credit risk.

3. Proposed Model:

3.1. Dataset Overview:

3.1.1. Data Summary

- **Total Row:** Our dataset consists of 45,528 rows in total, out of which we have made use of a set of 30,000 records to test our model to keep a holistic approach, while also ensuring a manageable set for data analysis.
- **Columns:** There are a total of 19 columns present in our dataset, that comprise a wide range of attributes from features like demography, financial background, employment information, and behavioral patterns.
- **Target Variable:** 'credit_card_default' is our target variable for the project, where it is defined by binary values to indicate if a person has defaulted or not. '0' indicates a **no default**, while '1' stands for a **default.**

3.1.2. Feature Types

- **Categorical Features:** These features are the ones that describe details of the borrower's behavioral patterns, lifestyle, demography, and any risk indicators. These variables include gender, owns_car, owns_house, occupation_type, migrant_worker, and default_in_last_6months.
- **Numeric Features:** Quantifying the key aspects of the financial situation and credit behavior of the borrower, these attributes include factors like age, net_yearly_income, credit_limit, credit_limit_used(%), etc.

3.1.3. Key Columns

- **Demographics:** These features provide an idea about the borrower's stability and spending patterns, which are indicative of their likelihood to default. These include age, gender, owns_car, owns_house, and no_of_children.
- **Financial Features:** These attributes are pivotal in understanding credit utilization and overall financial standing. The features includenet_yearly_income, credit_limit, credit_limit_used(%), and yearly_debt_payments.
- **Employment:** Features like no_of_days_employed and occupation_type determine the job stability and income potential of the individual.
- **Behavior & Risk Indication:** Features like prev_defaults, default_in_last_6months, and credit_score are used to check for any red flags in the borrower's pass, highlighting patterns and determining potential risk involved.

3.2. Data Pre-Processing:

Before applying machine learning models to predict credit risk, the data undergoes a series of preprocessing steps to ensure its quality, consistency, and compatibility with the modeling process. These steps involve handling missing or incomplete data, transforming categorical variables into numeric formats, scaling numerical features for uniformity, and addressing issues such as class imbalance. Additionally, creating new features based on domain knowledge can enhance the model's performance by providing more relevant information. Below are the specific preprocessing steps performed:

- **Handling missing values:** The dataset being used is first checked for any inconsistent or missing values in critical fields to ensure data quality and reliability. Any issues were addressed through imputation, using domain-specific strategies to maintain data integrity.
- **Encoding categorical variables:** Techniques like Label Encoding for ordinal data and One-Hot Encoding for nominal data were then applied to transform categorical variables into binary format, helping them become compatible with the machine learning algorithms being used.
- **Scaling numeric attributes:** Min-Max Scaling was applied to numeric variables, such as age and credit score, ensuring all model inputs were on a similar scale and preventing bias. This was essential to ensure that models sensitive to feature scaling performed optimally.
- **Addressing class imbalance:** The Synthetic Minority Oversampling Technique (SMOTE) was used to generate synthetic samples for the minority class, helping balance the dataset and improving model accuracy in predicting rare events like defaults.
- **Deriving additional features:** New features, including the credit-to-income ratio, were created to enhance both the interpretability and predictive power of the models.

This preprocessing pipeline ensures that the data is ready for modeling and increases the likelihood of producing accurate and reliable results.

3.3. Machine Learning Models Used:

After an iterative process of selecting the most suitable models for credit risk assessment, we narrowed our choices down to the following machine-learning techniques:

- **Logistic Regression**: As a fundamental binary classifier, logistic regression offers simplicity and interpretability, making it easy to understand predictions regarding borrower defaults.
- **K-Nearest Neighbors (KNN)**: Leveraging proximity-based similarity measures, KNN is effective in identifying patterns based on data points that are similar, making it a valuable model for credit risk assessment.
- **Decision Trees:** Using a tree-based structure, this follows a simplistic approach to identify the key attributes influencing credit risk while providing good performance on smaller datasets.

- **Random Forests**: This model is crucial in credit risk analysis as it helps identify significant features in the dataset while mitigating overfitting through an ensemble method built using multiple decision trees, improving generalization through continuous averages of individual trees, thus enhancing robustness and producing reliable results.
- **Gradient Boosting Machines (GBMs)**: GBMs excel in identifying complex, non-linear relationships within the data, which is essential for uncovering hidden patterns and improving predictive accuracy in credit risk models.
- **XGBoost:** An optimized implementation of gradient boosting, XGBoost offers faster computation and enhanced accuracy by utilizing advanced regularization techniques. It is particularly effective for large datasets with diverse features.

These models were carefully selected for their ability to handle various aspects of credit risk prediction, from simplicity and interpretability to the ability to capture complex patterns in the data.

3.4. Evaluation Metrics:

The ML models used in our assessment are quite different when it comes to aspects of functionality and overall usage, hence we needed to design the key metrics based on the common comparative factors to help understand which factors stood out and were the most relevant in our overall evaluation. The evaluation process involves cross-validation to ensure the robustness and reliability of results. The following metrics help analyze the overall prediction quality of the participating models in our research:

- **Accuracy**: Focusing on the model's ability to make predictions of the defaulters and non-defaulters, this metric is measured by calculating the percentage of accurate predictions made from the total instances.
- **Precision**: While accuracy focuses on both defaulters and non-defaulters, precision is a metric that tests the model's ability to predict the true defaults from the total predicted defaults, and its ability to minimize cases of false positives.
- **Recall**: Recall assesses the model's ability to identify all actual defaults, emphasizing the identification of true positives, even at the risk of including false positives.
- **F1 Score**: The F1 Score is the harmonic mean of precision and recall, balancing the two metrics to provide a single value that reflects both the model's ability to correctly identify defaults and avoid false positives.
- **Area Under the Receiver Operating Characteristic Curve (AUC-ROC)**: This metric evaluates the model's ability to discriminate between classes, considering all classification thresholds. A higher AUC indicates better model performance in distinguishing between defaults and non-defaults.

4. Results:

4.1. Descriptive Analysis:

Figure 1 illustrates the distribution of defaulters and non-defaulters in the test dataset, highlighting a notable class imbalance. Defaulters account for only 8%, while non-defaulters make up 92%, emphasizing the importance of selecting appropriate evaluation metrics like precision and recall.

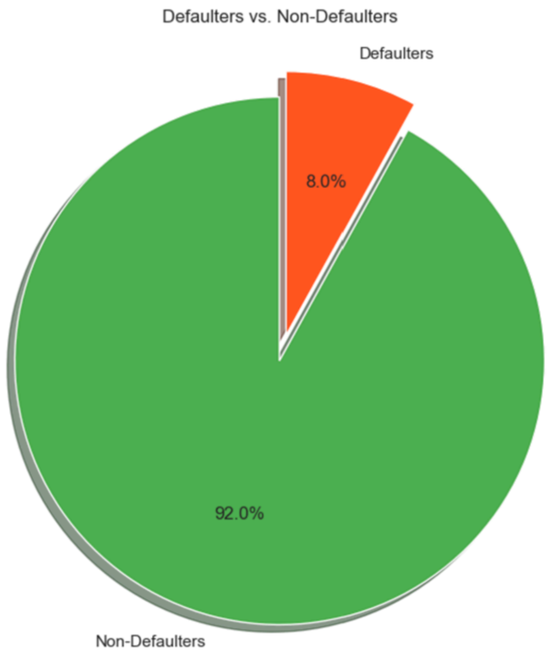

Figure 1: Pie-Chart for Defaulters & Non-Defaulters from Test Data

Figure 2 examines the relationship between occupation type, net yearly income, and credit default rates, revealing varying income levels across occupations. Notably, laborers display significant income variability.

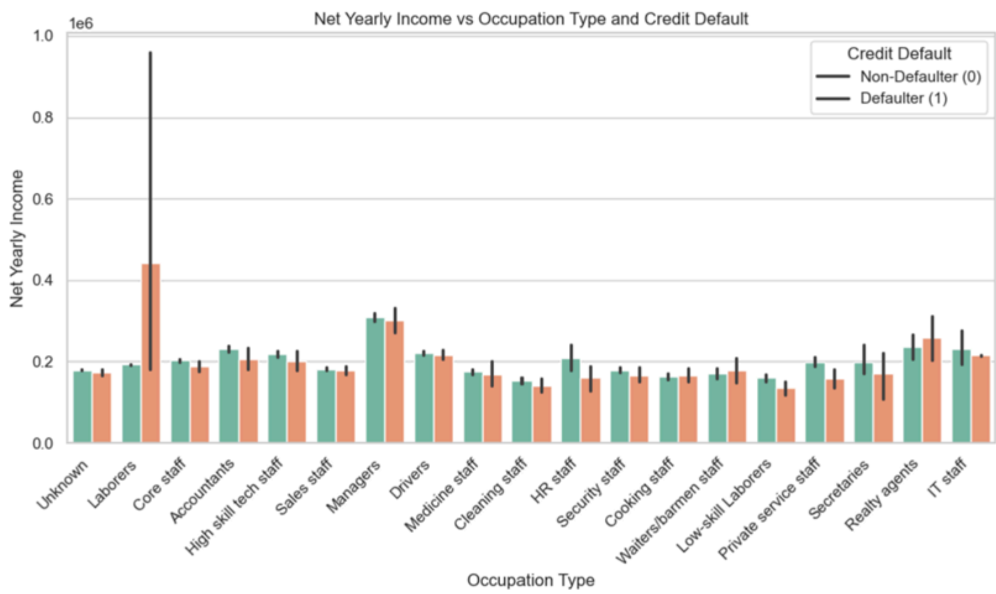

Figure 2: Bar Plot for Net Yearly Income vs Occupation Type & Credit Default

Figure 3 depicts the credit score distribution, demonstrating that non-defaulters generally have higher and more consistent scores compared to defaulters, underscoring the predictive relevance of credit scores.

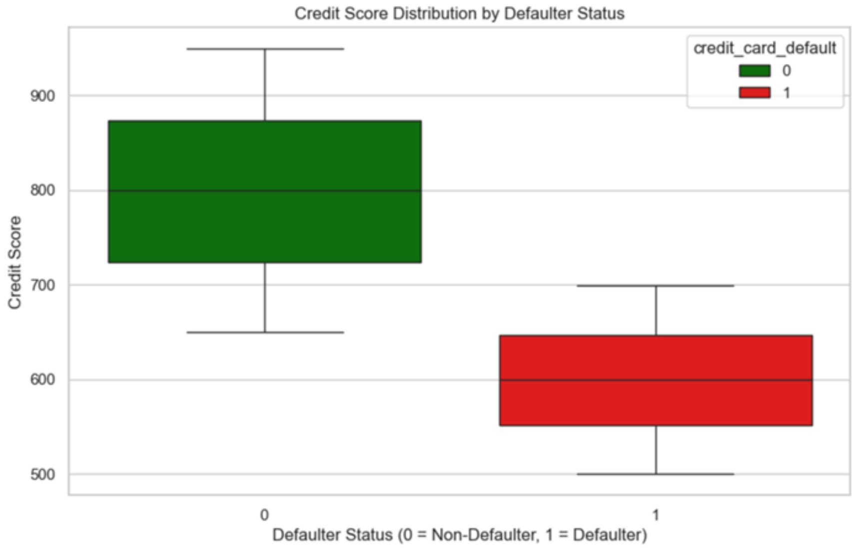

Figure 3: Credit Score Distribution by Defaulter Status

Figure 4 explores credit limit utilization and default likelihood, showing that defaulters tend to utilize a higher percentage of their credit limit, which proves a key predictor of default.

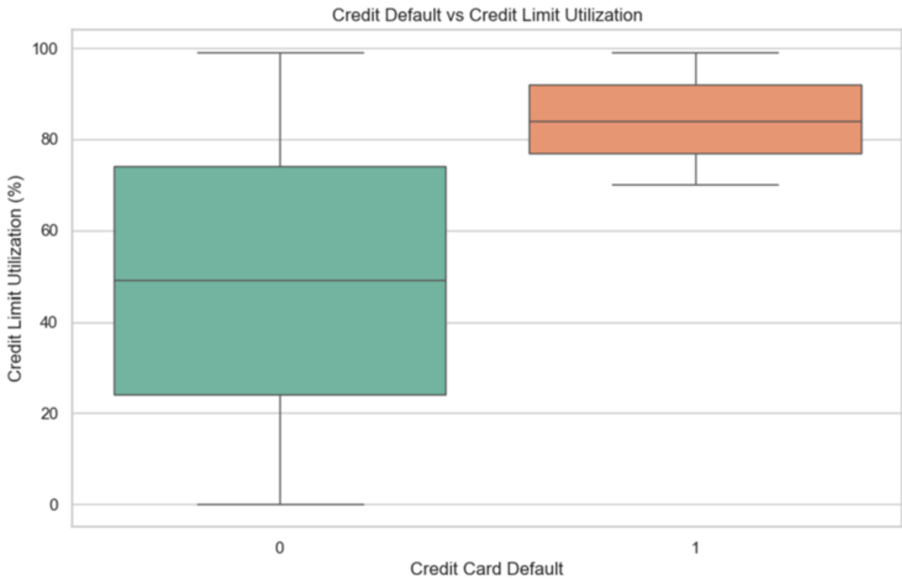

Figure 4: Credit Default vs Credit Limit Utilization

Figure 5 presents gender distribution in the dataset, indicating that females constitute the largest group, with default rates being lower across all genders.

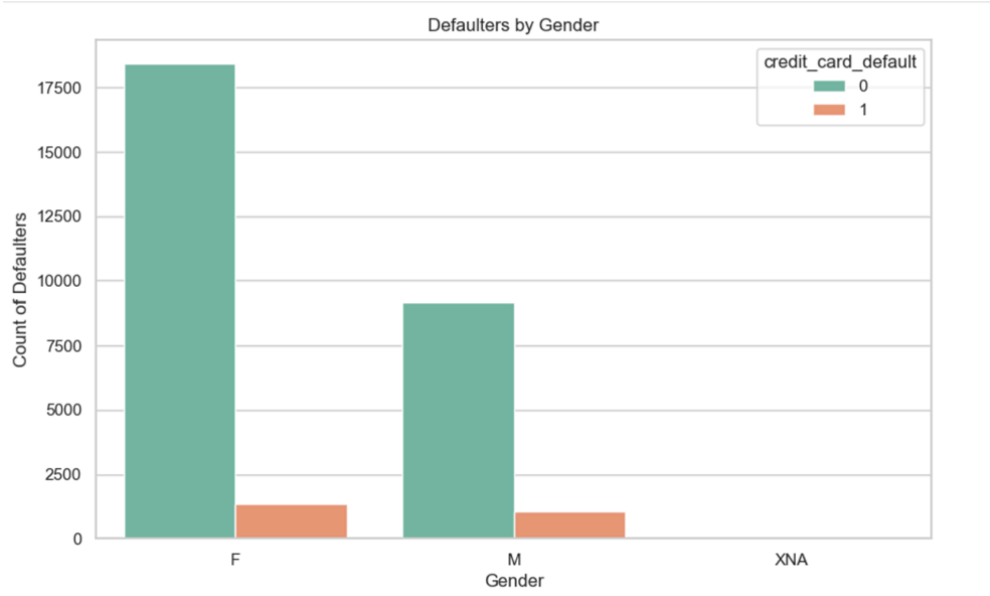

Figure 5: Count of Defaulters vs Defaulters by Gender

Figure 6 compares the distribution of continuous variables such as income and credit score, revealing data variability and identifying outliers that could impact model performance. While most data points fall within expected ranges, certain variables display significant extremes.

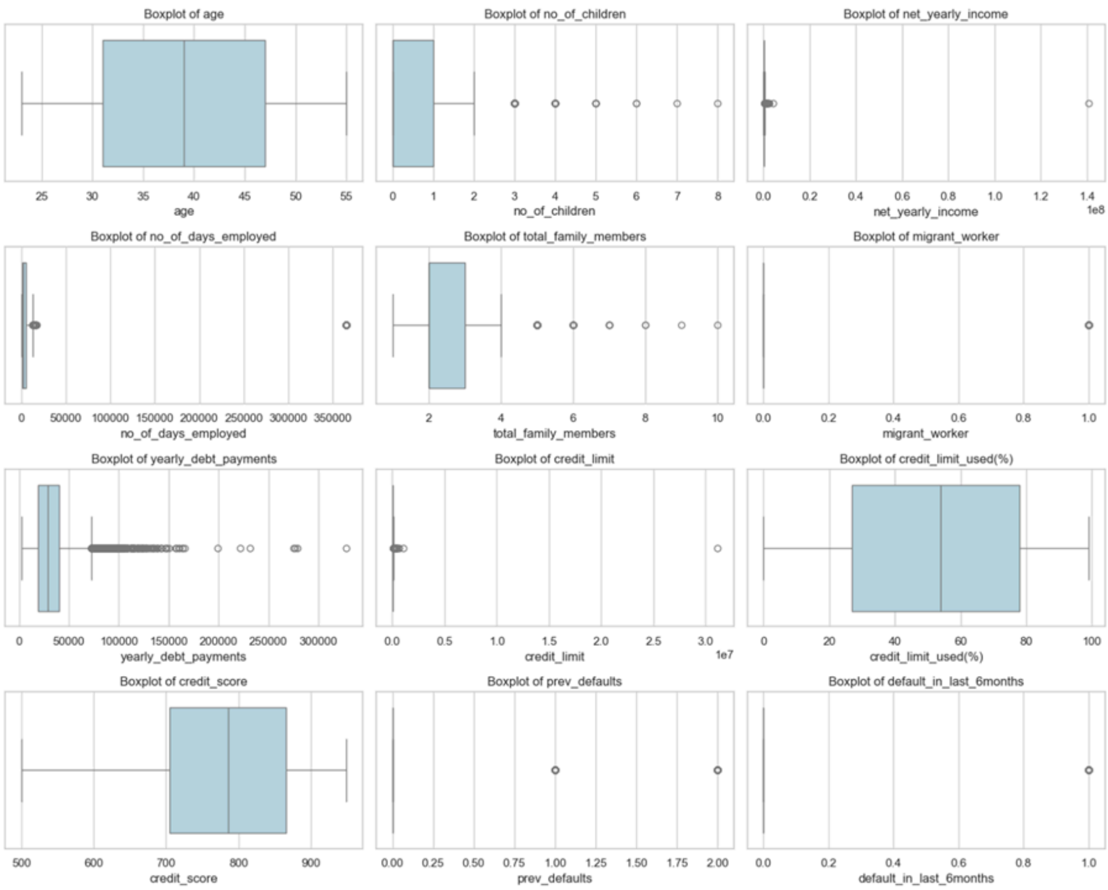

Figure 6: Boxplot for all Continuous variables

4.2. Comparative Analysis:

4.2.1. Logistic Regression

Accuracy Score (for Train Data): 95.74081

Accuracy Score (for Test Data): 94.66666

F1_Score (for Train Data): 95.0731

F1_Score (for Test Data): 86.00746

RMSE: 0.23094

ROC AUC Score: 0.95924

Logisitc Regression

```
[376]: # initialize LogisticRegression and fit the model
       logreg = LogisticRegression()
       logreg.fit(X_train, y_train)

[376]: ▾ LogisticRegression
       LogisticRegression()

[377]: # prediction
       evaluation(logreg)
```

```
===================================================
[[18245  1031]
 [  611 18665]]
              precision    recall  f1-score   support

           0       0.97      0.95      0.96     19276
           1       0.95      0.97      0.96     19276

    accuracy                           0.96     38552
   macro avg       0.96      0.96      0.96     38552
weighted avg       0.96      0.96      0.96     38552

Accuracy of TRAIN data: 95.74081759701183
F1_Score of TRAIN data: 95.74031202556064
===================================================
[[7800  461]
 [  19  720]]
              precision    recall  f1-score   support

           0       1.00      0.94      0.97      8261
           1       0.61      0.97      0.75       739

    accuracy                           0.95      9000
   macro avg       0.80      0.96      0.86      9000
weighted avg       0.97      0.95      0.95      9000

Accuracy of TEST data: 94.66666666666667
F1_Score of TEST data: 86.00746268656717
===================================================
RMSE:  0.23094010767585033
ROC AUC score:  0.9592425992390677
```

Figure 7: Predictive Analysis Score for Logistic Regression

4.2.2. Decision Trees

Accuracy Score (for Train Data): 100.00

Accuracy Score (for Test Data): 96.77777

F1_Score (for Train Data): 100.00

F1_Score (for Test Data): 90.05335

RMSE: 0.17950

ROC AUC Score: 0.93069

Decision Tree

```
[379]: tree_model = DecisionTreeClassifier()
       tree_model.fit(X_train, y_train)

[379]: ▾ DecisionTreeClassifier
       DecisionTreeClassifier()

[380]: # prediction
       evaluation(tree_model)
```

```
==================================================
[[19276     0]
 [    0 19276]]
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     19276
           1       1.00      1.00      1.00     19276

    accuracy                           1.00     38552
   macro avg       1.00      1.00      1.00     38552
weighted avg       1.00      1.00      1.00     38552

Accuracy of TRAIN data: 100.0
F1_Score of TRAIN data: 100.0
==================================================
[[8055  206]
 [  84  655]]
              precision    recall  f1-score   support

           0       0.99      0.98      0.98      8261
           1       0.76      0.89      0.82       739

    accuracy                           0.97      9000
   macro avg       0.88      0.93      0.90      9000
weighted avg       0.97      0.97      0.97      9000

Accuracy of TEST data: 96.77777777777777
F1_Score of TEST data: 90.05335365853658
==================================================
RMSE:  0.17950549357115014
ROC AUC score:  0.9306982169507373
```

Figure 8: Predictive Analysis Score for Decision Trees

4.2.3. Random Forests

Accuracy Score (for Train Data): 100.00
Accuracy Score (for Test Data): 96.5
F1_Score (for Train Data): 100.00
F1_Score (for Test Data): 89.50838
RMSE: 0.18708
ROC AUC Score: 0.93780

Random Forest

```
[382]: regr_rfr = RandomForestClassifier(random_state=42, oob_score=True)
       regr_rfr.fit(X_train, y_train)
```

```
[382]:  RandomForestClassifier
        RandomForestClassifier(oob_score=True, random_state=42)
```

```
[383]: # prediction
       evaluation(regr_rfr)
```

```
==================================================
[[19276     0]
 [    0 19276]]
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     19276
           1       1.00      1.00      1.00     19276

    accuracy                           1.00     38552
   macro avg       1.00      1.00      1.00     38552
weighted avg       1.00      1.00      1.00     38552

Accuracy of TRAIN data: 100.0
F1_Score of TRAIN data: 100.0
==================================================
[[8016  245]
 [  70  669]]
              precision    recall  f1-score   support

           0       0.99      0.97      0.98      8261
           1       0.73      0.91      0.81       739

    accuracy                           0.96      9000
   macro avg       0.86      0.94      0.90      9000
weighted avg       0.97      0.96      0.97      9000

Accuracy of TEST data: 96.5
F1_Score of TEST data: 89.50838979096383
==================================================
RMSE:  0.18708286933869708
ROC AUC score:  0.9378099877163822
```

Figure 9: Predictive Analysis Score for Random Forest

4.2.4. Light GBM

Accuracy Score (for Train Data): 99.11029

Accuracy Score (for Test Data): 96.67777

F1_Score (for Train Data): 99.11024

F1_Score (for Test Data): 90.041296

RMSE: 0.18226

ROC AUC Score: 0.94370

LightGBM

```
[394]: lgbm_model = LGBMClassifier(random_state=42)
       lgbm_model.fit(X_train, y_train)

[LightGBM] [Warning] Found whitespace in feature_names, replace with underlines
[LightGBM] [Info] Number of positive: 19276, number of negative: 19276
[LightGBM] [Info] Auto-choosing row-wise multi-threading, the overhead of testing was 0.002016 seconds.
You can set `force_row_wise=true` to remove the overhead.
And if memory is not enough, you can set `force_col_wise=true`.
[LightGBM] [Info] Total Bins 932
[LightGBM] [Info] Number of data points in the train set: 38552, number of used features: 24
[LightGBM] [Info] [binary:BoostFromScore]: pavg=0.500000 -> initscore=0.000000

[394]: LGBMClassifier
       LGBMClassifier(random_state=42)

[395]: # prediction
       evaluation(lgbm_model)

==================================================
[[18968   308]
 [   35 19241]]
              precision    recall  f1-score   support

           0       1.00      0.98      0.99     19276
           1       0.98      1.00      0.99     19276

    accuracy                           0.99     38552
   macro avg       0.99      0.99      0.99     38552
weighted avg       0.99      0.99      0.99     38552

Accuracy of TRAIN data: 99.11029259182403
F1_Score of TRAIN data: 99.11024797481507
==================================================
[[8024  237]
 [  62  677]]
              precision    recall  f1-score   support

           0       0.99      0.97      0.98      8261
           1       0.74      0.92      0.82       739

    accuracy                           0.97      9000
   macro avg       0.87      0.94      0.90      9000
weighted avg       0.97      0.97      0.97      9000

Accuracy of TEST data: 96.67777777777778
F1_Score of TEST data: 90.0412969761847
==================================================
RMSE:  0.18226964152656422
ROC AUC score:  0.9437069104891351
```

Figure 9: Predictive Analysis Score for LightGBM

4.2.5. XGBoost

Accuracy Score (for Train Data): 99.42415

Accuracy Score (for Test Data): 96.93333

F1_Score (for Train Data): 99.42414

F1_Score (for Test Data): 90.63899

RMSE: 0.18226

ROC AUC Score: 0.94370

XGBoost

```
[388]: xgb_model = XGBClassifier(random_seed=42)
       xgb_model.fit(X_train, y_train)
```

```
C:\Users\mohak\anaconda3\Lib\site-packages\xgboost\core.py:158: UserWarning: [16:32:40] WARNING: C:\
caling-group-i-0c55ff5f71b100e98-1\xgboost\xgboost-ci-windows\src\learner.cc:740:
Parameters: { "random_seed" } are not used.

  warnings.warn(smsg, UserWarning)
```

```
[388]:
XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric=None, feature_types=None,
              gamma=None, grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=None, max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=None, max_leaves=None,
              min_child_weight=None, missing=nan, monotone_constraints=None,
              multi_strategy=None, n_estimators=None, n_jobs=None,
              num_parallel_tree=None, random_seed=42, ...)
```

```
[389]: # prediction
       evaluation(xgb_model)
```

```
================================================
[[19083   193]
 [   29 19247]]
              precision    recall  f1-score   support

           0       1.00      0.99      0.99     19276
           1       0.99      1.00      0.99     19276

    accuracy                           0.99     38552
   macro avg       0.99      0.99      0.99     38552
weighted avg       0.99      0.99      0.99     38552

Accuracy of TRAIN data: 99.42415438887735
F1_Score of TRAIN data: 99.4241439679159
================================================
[[8052  209]
 [  67  672]]
              precision    recall  f1-score   support

           0       0.99      0.97      0.98      8261
           1       0.76      0.91      0.83       739

    accuracy                           0.97      9000
   macro avg       0.88      0.94      0.91      9000
weighted avg       0.97      0.97      0.97      9000

Accuracy of TEST data: 96.93333333333334
F1_Score of TEST data: 90.63899063899063
================================================
RMSE:  0.17511900715418263
ROC AUC score:  0.9420186706403191
```

Figure 10: Predictive Analysis Score for XGBoost

4.2.6. K-Nearest Neighbors(KNN)

Accuracy Score (for Train Data): 99.36968

Accuracy Score (for Test Data): 96.81111

F1_Score (for Train Data): 99.36965

F1_Score (for Test Data): 90.04947

RMSE: 0.17857

ROC AUC Score: 0.92595

Choose k = 2

```
[568]: # # building knn and fit the model
knn = KNeighborsClassifier(n_neighbors = 2)
knn.fit(X_train, y_train)
```

```
[568]: KNeighborsClassifier
KNeighborsClassifier(n_neighbors=2)
```

```
[569]: # # Evaluation

evaluation(knn)
```

```
==================================================
[[19276     0]
 [  243 19033]]
              precision    recall  f1-score   support

           0       0.99      1.00      0.99     19276
           1       1.00      0.99      0.99     19276

    accuracy                           0.99     38552
   macro avg       0.99      0.99      0.99     38552
weighted avg       0.99      0.99      0.99     38552

Accuracy of TRAIN data: 99.36968250674414
F1_Score of TRAIN data: 99.36965746322618
==================================================
[[8066  195]
 [  92  647]]
              precision    recall  f1-score   support

           0       0.99      0.98      0.98      8261
           1       0.77      0.88      0.82       739

    accuracy                           0.97      9000
   macro avg       0.88      0.93      0.90      9000
weighted avg       0.97      0.97      0.97      9000

Accuracy of TEST data: 96.81111111111112
F1_Score of TEST data: 90.04947870709792
==================================================
RMSE:  0.17857460314638499
ROC AUC score:  0.9259512760203765
```

Figure 10: Predictive Analysis Score for KNN

4.2.7. Comparison Table

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	0.9466	0.80	0.96	0.86	0.9592
Decision Tree	0.9677	0.88	0.93	0.90	0.9306
Random Forest	0.9650	0.86	0.94	0.90	0.9378
LightGBM	0.9667	0.87	0.94	0.90	0.9437
XGBoost	0.9693	0.88	0.94	0.91	0.9420
KNN	0.9681	0.88	0.93	0.90	0.9259

Based on the above analysis, we identified XGBoost as the most suitable model for our credit risk assessment project. It achieved the highest accuracy (96.93%) and F1-Score (0.91), demonstrating robust performance across multiple evaluation metrics. Its optimized gradient-boosting algorithm ensures computational efficiency while maintaining excellent predictive accuracy.

4.3. Feature Importance

XGBoost highlighted critical features influencing credit risk, such as:

- Credit-to-income ratio.
- Historical default records.
- Utilization percentage of credit limits.

Other features, like gender, occupation_Laborers, and occupation_Unknown, have lower F scores, indicating lesser importance in the model.

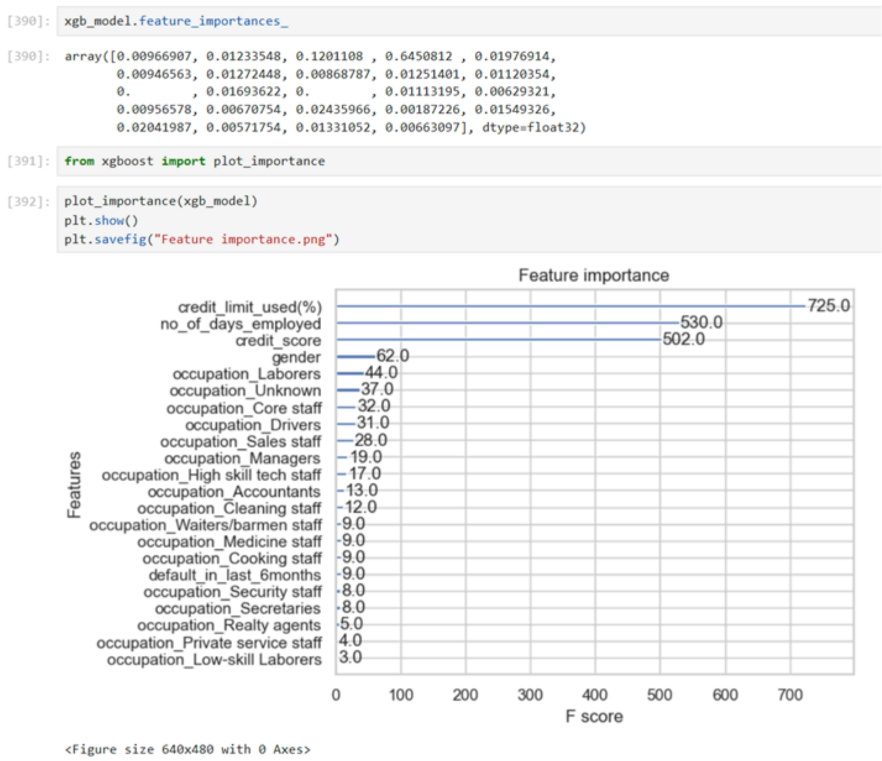

Figure 11: Feature Importance

5. Discussion:

The proposed methodology demonstrated considerable efficacy, highlighting the superiority of machine learning (ML) models over traditional techniques for credit risk evaluation. Specifically, **XGBoost** emerged as a strong performer due to its ability to handle intricate data structures with precision and efficiency. Meanwhile, **Decision Trees** and **Random Forests** offered a balanced approach, excelling in interpretability and predictive accuracy, making them highly dependable in scenarios requiring explainable results.

Nevertheless, several challenges were encountered:

- **Optimization challenges**: Managing the high computational cost associated with models like XGBoost necessitated the development and implementation of cost-saving optimization strategies.
- **Data pre-processing requirements**: Significant investment was required to address critical issues such as missing data and class imbalance, emphasizing the need for a robust preprocessing pipeline.

These findings underscore the critical importance of selecting ML models that align with the specific requirements of the task at hand. Whether the focus is on **accuracy**, **interpretability**, or **computational**

efficiency, the methodology highlights the need for a strategic approach to model selection to ensure optimal performance in varying application contexts.

6. Conclusion:

With this project, we effectively demonstrated the potential of machine learning (ML) models for credit risk assessment, with **XGBoost** standing out as the most efficient and accurate model. By employing advanced techniques and rigorous evaluation metrics, the study achieved a robust framework for predicting credit defaults.

- **XGBoost**: Delivered the highest accuracy and efficiency, making it particularly suitable for large-scale and complex datasets.
- **Random Forests**: Struck a balance between predictive performance and interpretability, making it valuable for feature importance analysis.
- **Derived Features**: Enhanced model performance significantly, with metrics like the credit-to-income ratio proving critical for accuracy improvements.

These findings not only confirm the applicability of advanced ML models in financial domains but also emphasize the importance of feature engineering and context-specific model selection for achieving optimal results.

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